Xiaodong Huang

huan2814@umn.edu | Personal Website | LinkedIn | GitHub

EDUCATION

University of Minnesota Twin Cities (UMN) Minneapolis, MN

B.S. in Statistical Science; Minor in Computer Science *Sep. 2023 – May 2026*

- **Cumulative GPA**: 3.89/4.00; **Statistics and Computer Science GPA**: 3.94/4.00

RESEARCH INTERESTS

Statistical machine learning; clinical prediction and decision-making; causal inference and real-world evidence; knowledge-centric methods and trustworthy clinical AI systems.

PUBLICATIONS & MANUSCRIPTS

- Jin, Y., **Huang, X.**, Wang, Y., Wacker, D. A., Puskarich, M. A., and Xie, F. *Development and Multicenter External Validation of a Data-Driven Scoring System for Early and Rapid Identification of Sepsis in Emergency Departments.* **Accepted for publication** in *eClinicalMedicine*. medRxiv preprint.
- **Huang, X.**, Patel, S., Jin, Y., Mai, X., Chu, C., Hou, J., Puskarich, M. A., and Xie, F. (2026). *ActionRef: Learning Reference Thresholds and Guiding Clinical Risk Threshold Selection.* Submitted to *Journal of the American Medical Informatics Association (JAMIA)*.

CONFERENCE & WORKSHOP PAPERS

- Jin, Y.*, **Huang, X.***, Chandu, H. P., Xiong, X., Lin, M., and Xie, F. (2026). *RADAR: A Guideline-Grounded Multi-Agent System for the Emergency-Department Sepsis Pathway.* Accepted to the *NeurIPS 2026 Workshop on GenAI4Health*, Demonstration Paper Track. *Co-first authors. OpenReview.

RESEARCH EXPERIENCE

University of Minnesota Twin Cities *May 2025 – Present*

Supervisor: **Dr. Feng Xie**, Assistant Professor, Division of Computational Health Sciences

Clinical Machine Learning and Decision Systems for Emergency Care

- **ActionRef:** Developed a general framework that translates observed clinical actions into interpretable risk thresholds and guides selection under workload constraints; led multicenter evaluation and released a reusable Python package.
- **RADAR:** Co-developed a guideline-grounded multi-agent system for clinical decision support, integrating clinical knowledge graphs and a shared patient state; built core patient-data and evidence-integration workflows and contributed to the public software package and interactive demo.
- **Sepsis risk prediction:** Co-developed an interpretable sepsis risk score using more than 2.2 million ED encounters; harmonized cohorts and implemented model-development and external-validation workflows across UMN, MIMIC-IV-ED, and Stanford MC-MED.
- **LLM-Augmented AutoScore:** Developed reusable data-processing and LLM feature-extraction pipelines for integrating clinical narratives with interpretable risk scores; implemented comparative clinical benchmarking of AutoScore models against established sepsis scores.

Peking University *May 2026 – Present*

Research collaboration with **Dr. Shenda Hong**, Assistant Professor, National Institute of Health Data Science

Causal Inference and Dynamic Patient Modeling in Critical Care

- **Real-world treatment evidence:** Designed and implemented multicenter target trial emulations of hydrocortisone initiation during vasopressor support; constructed treatment-exposure cohorts and estimated adjusted associations with hospital mortality in MIMIC-IV and eICU.
- **Dynamic patient modeling:** Developed statistical and deep learning models of evolving ICU physiology; built a cross-hospital evaluation framework for multi-step forecasting, uncertainty quantification, and model transfer between MIMIC-IV and eICU.

RESEARCH EXPERIENCE (CONTINUED)

National University of Singapore *Sep. 2026 – Present*

Research Assistant, Yu Liu Lab, Department of Biomedical Engineering

- Developing knowledge-centric approaches to trustworthy clinical AI, connecting guideline-based reasoning, evidence verification, and sequential decision-making under incomplete and evolving patient information.
- Investigating adaptive information acquisition and uncertainty-aware recommendations, including when a system should seek additional evidence, revise a decision, or defer to clinical judgment.

University of Minnesota Twin Cities *Jul. 2025 – Oct. 2025*

Research in the group of **Dr. Jie Ding**, Associate Professor, School of Statistics

AI and Data Science for Collaborative Research and Life Sciences

- Explored LLM agents for collaborative scientific research and biomedical knowledge extraction under the guidance of Ph.D. student Jin Du.
- Tested agent-assisted workflows for document analysis, scientific writing, and visualization; contributed to technical discussions and prototype evaluation.

PRESENTATIONS

- **Huang, X.**, Patel, S., Jin, Y., Mai, X., and Xie, F. (2026). *From Risk Prediction to Clinical Action: A Universal Behavior-Informed Framework for Learning and Governing Clinical Decision Thresholds from Real-World Practice*. Oral presentation at the University of Minnesota Undergraduate Research Symposium.
- Jin, Y., **Huang, X.**, Wang, Y., Wacker, D. A., Puskarich, M., and Xie, F. (2025). *Development and Multicenter External Validation of a Data-Driven Scoring System for Early and Rapid Identification of Sepsis in Emergency Departments*. Co-authored work; oral presentation at the **AMIA 2025 Annual Symposium**.

HONORS & AWARDS

- **2026 Love of Learning Grant**, The Phi Beta Kappa Society — Jun. 2026
- **Elected Member**, The Phi Beta Kappa Society — May 2026
- **UROP Research Award (competitive)**, University of Minnesota (Spring 2026) — Dec. 2025
- **Dean's List**, University of Minnesota — 2023–2026

ADDITIONAL PROFESSIONAL EXPERIENCE

Hubei Hanjiang Guotou Digital Industry Group Co., Ltd. *Jun. 2025 – Dec. 2025*

Data Analyst Intern

- Built SQL/Python pipelines that cleaned and standardized more than 60,000 IoT and municipal records for a city-management platform and supported LLM-assisted retrieval over operational data.

Shenzhen Manjinghua Investment Group *May 2024 – Aug. 2024*

R&D Intern, Big Data R&D Center

- Analyzed multi-source operational data and developed regression-based foot-traffic forecasts and dashboards in Python and MATLAB to support resource allocation and predictive-maintenance analyses.

METHODS & TECHNICAL SKILLS

Programming	Python, R, SQL, Stata, Java, MATLAB
Libraries	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, pandas, NumPy
Tools	Git, LaTeX
Statistics	Causal inference, target trial emulation, longitudinal modeling, survival analysis, uncertainty quantification, predictive modeling
AI Methods	Machine learning, deep learning, natural language processing, retrieval-augmented generation (RAG), knowledge graphs, multi-agent systems

SELECTED COURSEWORK

Statistical Theory I & II; Regression and Correlated Data; Statistical Machine Learning I & II; Applied Multivariate Methods; Nonparametric Methods; Data Structures and Algorithms; Computational Linear Algebra.